

NANDHAKUMAR M

Full Stack AI Engineer

Coimbatore, Tamil Nadu, India · Open to relocation to Canada

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PROFESSIONAL SUMMARY

Full stack software engineer with 5+ years of experience building enterprise web applications, data platforms and Generative AI solutions for healthcare, business continuity and analytics. Skilled in Python (FastAPI, Django), React, Node.js, PostgreSQL, Azure and AWS. Recent work includes LLM-powered applications with OpenAI and Ollama, AI agents that call tools through the Model Context Protocol, and document intelligence with OCR. Recognized as Best Performer at Firstsource Solutions.

TECHNICAL SKILLS

AI & GenAI: OpenAI API, Azure OpenAI, Ollama, LLM tool calling, Model Context Protocol (MCP), prompt engineering, sentiment analysis, OCR (Tesseract)

Backend: Python (FastAPI, Django REST Framework), Node.js, Express, Java (Spring Boot), REST APIs, microservices

Frontend: React, Next.js, TypeScript, JavaScript, React Native (Expo), Redux Toolkit, Tailwind CSS

Data: PostgreSQL, MySQL, MongoDB, Redis, PySpark, Microsoft Fabric Lakehouse, Azure Data Lake Storage

Cloud & DevOps: Azure, AWS, Docker, CI/CD, Azure DevOps, Git

Practices: JWT, OAuth 2.0 and SSO, automated testing (pytest, Vitest, Playwright), Agile delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Apr 2025 – Present

Firstsource Solutions Limited, Hyderabad, India

• Develop features for the GigSource project end to end, with React front ends and Node.js APIs.

• Build and maintain Azure DevOps CI/CD pipelines that build and deploy the application.

• Developed Python (PySpark) data pipelines on Microsoft Fabric Lakehouse in the first months of the role, with Azure Data Lake Storage for ingestion and archival.

• Reduced ETL processing time by **40%** and built dashboards that improved data accessibility and business insight by **35%**.

Junior Software Engineer

Oct 2021 – Mar 2025

Firstsource Solutions Limited, Hyderabad, India

• Led full-cycle development of Information Lifecycle Management, Business Continuity Management and healthcare (NGO) platforms with **100%** on-time delivery.

• Developed Spring Boot microservices and React applications, improving system efficiency by **35%**.

• Reduced infrastructure costs by **25%** by optimizing AWS and Azure resource usage.

• Improved query performance by **30%** by redesigning PostgreSQL schemas.

• Automated Docker deployments to AWS (**40%** faster) and streamlined CI/CD for a **50%** faster release cycle.

• Wrote API and architecture documentation that reduced new-developer onboarding time by **40%**.

Full Stack Developer

Mar 2021 – Sep 2021

Invious Technologies, Coimbatore, India

• Led a cross-functional team and increased project delivery speed by **35%** through clear planning.

• Optimized the Hyper App project, improving system performance by **30%**.

• Built Spring Boot REST APIs that reduced API response time by **25%**.

• Managed AWS deployments and cut infrastructure costs by **15%**.

• Integrated MySQL and MongoDB, improving data retrieval speed by **40%**.

Career break

COVID-19 pandemic

Jun 2020 – Feb 2021

Java Web Developer

Smartway Industrial Automation, Coimbatore, India

Dec 2019 – May 2020

â€¢ Developed a real-time monitoring dashboard with Java and JSP, improving user accessibility by **20%**.

â€¢ Improved application performance by **20%** through code-level optimization.

SELECTED PROJECTS github.com/Nandhu29696

AI Email Assistant | Python, FastAPI, PostgreSQL, Next.js, OpenAI, Azure Blob Storage

Classifies, prioritizes and summarizes Gmail and Outlook email with an LLM and sentiment analysis; validates senders and attachments, converts and merges documents into one PDF, and replies to the sender automatically.

Mediance Healthcare CRM | React, TypeScript, Django REST Framework, Celery, MySQL

Client CRM and patient portal with six user roles: lead pipeline, consultations, prescriptions and lab reports, email/WhatsApp/SMS notifications, and an English, Hindi and Tamil interface.

Fleet Manager | React, Redux Toolkit, Node.js, Express, Sequelize, MySQL

Phased rewrite of a legacy PHP fleet system: vehicles, drivers, fuel, maintenance and salaries for own and sub-vendor fleets, with document-expiry alerts, PDF quotations and reports.

JobAgent AI | Python, FastAPI, React, Ollama

Career assistant that scores resume-to-job match, generates tailored cover letters and tracks applications.

MCP Healthcare Appointment Agent | Node.js, Express, PostgreSQL, Ollama

AI agent that selects and calls typed tools (patient search, doctor availability, booking) to book appointments from natural-language requests.

Loan Document Extraction | Django REST Framework, PyMuPDF, Tesseract OCR, React

Extracts structured fields from scanned mortgage documents into a searchable loan tracker.

HealthCamp Management Platform | Node.js, Express, PostgreSQL, Azure Blob Storage

Platform for an NGO's community health camps: camp planning, patient records, bulk data import and reports. In production at hms.nirmaan.org.

CERTIFICATIONS & TRAINING

GitHub Copilot Certification (Exam GH-300), Microsoft

Valid 2026 – 2028

Develop AI Apps and Agents on Azure (AI-103T00-A course), Microsoft

Valid 2026 – 2027

EDUCATION

Master of Computer Applications (MCA)

2020

Hindusthan College of Engineering and Technology (Anna University), Coimbatore, India

Bachelor of Computer Applications (BCA)

2016

Hindusthan College of Arts and Science, Coimbatore, India

AWARDS

Best Performer, Firstsource Solutions: for technical excellence and contributions to project delivery

Agility and Humanness Award, Firstsource Solutions: for adaptability and collaboration on high-impact projects